WHEN THE FITTING IDEAL OF A BOUNDED COMPLEX WITH RESPECT TO A NONZERODIVISOR IS PRINCIPAL, THE SHIFTED COMPLEX IS LOCALLY FREE AND ITS DIFFERENTIAL SPLITS OFF

THE STACKS PROJECT AUTHORS

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Editorial note. The proof refers to the fitting ideal statement of this chapter and to the section in which the shifted complex is constructed; both are carried as navigation stubs with their roles recorded in the reasons.

Editorial note. The statement uses the notation eta_f for the shifted complex and the ideals I_i(M, f) for the Fitting ideals, both as printed by the author in the surrounding section; the preamble supplies the macros involved.

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Allowed standard prerequisites (editorial).

- The Fitting ideal I_i(M, f) is generated by the r_i by r_i minors of the map (f, d^i) and its powers of f vanish after localizing at f. used at proof lines 28060-28068. Supplies the generator g and lets the author read the rank of the map after localizing, which gives the vanishing of the higher minors.
- A finite module whose Fitting invariant is principal and which is defined by the minors of a map is finite locally free of the corresponding rank, and a finite projective module over a local ring is free. used at proof lines 28069-28078. Shows that the quotient Q is finite locally free of rank r_{i+1} and f-torsion free, which identifies the cokernel modulo f-power torsion.
- The shifted complex eta_f M has as its i-th term the kernel of the map (d^i, -1) from f^i M^i direct sum f^{i+1} M^{i+1} to f^i M^{i+1}. used at proof lines 28094-28098. Identifies the term of the shifted complex with

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the kernel in question and thereby with a direct summand of the finite free module.

Source reference. Original lemma 0F7M: sources/stacks-more-algebra/pinned-original-source.tex, source lines 1616–1616. The reference is applied at proof lines 28075-28076 to conclude that the quotient by g-torsion is finite locally free of rank r_{i+1}; the identification of the shifted term as a direct summand, which is the conclusion, is obtained inside the counted span.

Source reference. Original section 0F7N: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27385–27385. The reference points to the construction of the shifted complex, used at proof lines 28093-28096 to identify the image of the injective map with the kernel of the second differential; it is a pointer to the definition, not a substitute for the computation in the span.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a bounded complex of finite free A-modules. Assume $I_i(M^\bullet, f)$ is a principal ideal. Then $(\eta_f M)^i$ is locally free of rank r_i and the map $(1, d^i) : (\eta_f M)^i \to f^i M^i \oplus f^{i+1} M^{i+1}$ is the inclusion of a direct summand.*

Proof. Choose a generator g for $I_i(M^\bullet, f)$. Since $f^{r_i} \in I_i(M^\bullet, f)$ we see that g divides a power of f. In particular g is a nonzerodivisor in A. The $r_i \times r_i$-minors of the map $(f, d^i) : M^i \to M^i \oplus M^{i+1}$ generate the ideal $I_i(M^\bullet, f)$ and the $(r_i + 1) \times (r_i + 1)$-minors of (f, d^i) are zero: we may check this after localizing at f where the rank of the map is equal to r_i. Consider the surjection

$$M^i \oplus M^{i+1} \longrightarrow Q = \text{Coker}(f, d^i)/g\text{-torsion}$$

By Lemma 0F7M the module Q is finite locally free of rank r_{i+1}. Hence Q is f-torsion free and we conclude the cokernel of (f, d^i) modulo f-power torsion is Q as well.

Consider the complex of finite free A-modules

$$0 \to f^{i+1} M^i \xrightarrow{1, d^i} f^i M^i \oplus f^{i+1} M^{i+1} \xrightarrow{d^i, -1} f^i M^{i+1} \to 0$$

which becomes split exact after localizing at f. The map $(1, d^i) : f^{i+1} M^i \to f^i M^i \oplus f^{i+1} M^{i+1}$ is isomorphic to the map $(f, d^i) : M^i \to M^i \oplus M^{i+1}$ we studied above. Hence the image

$$Q' = \text{Im}(f^i M^i \oplus f^{i+1} M^{i+1} \xrightarrow{d^i, -1} f^i M^{i+1})$$

is isomorphic to Q in particular projective. On the other hand, by construction of η_f in Section 0F7N the image of the injective map $(1, d^i) : (\eta_f M)^i \to f^i M^i \oplus f^{i+1} M^{i+1}$ is the kernel of $(d^i, -1)$. We conclude that we obtain an isomorphism $(\eta_f M)^i \oplus Q' = f^i M^i \oplus f^{i+1} M^{i+1}$ and we see that indeed $\eta_f M^i$ is finite locally free of rank r_i and that $(1, d^i)$ is the inclusion of a direct summand. □